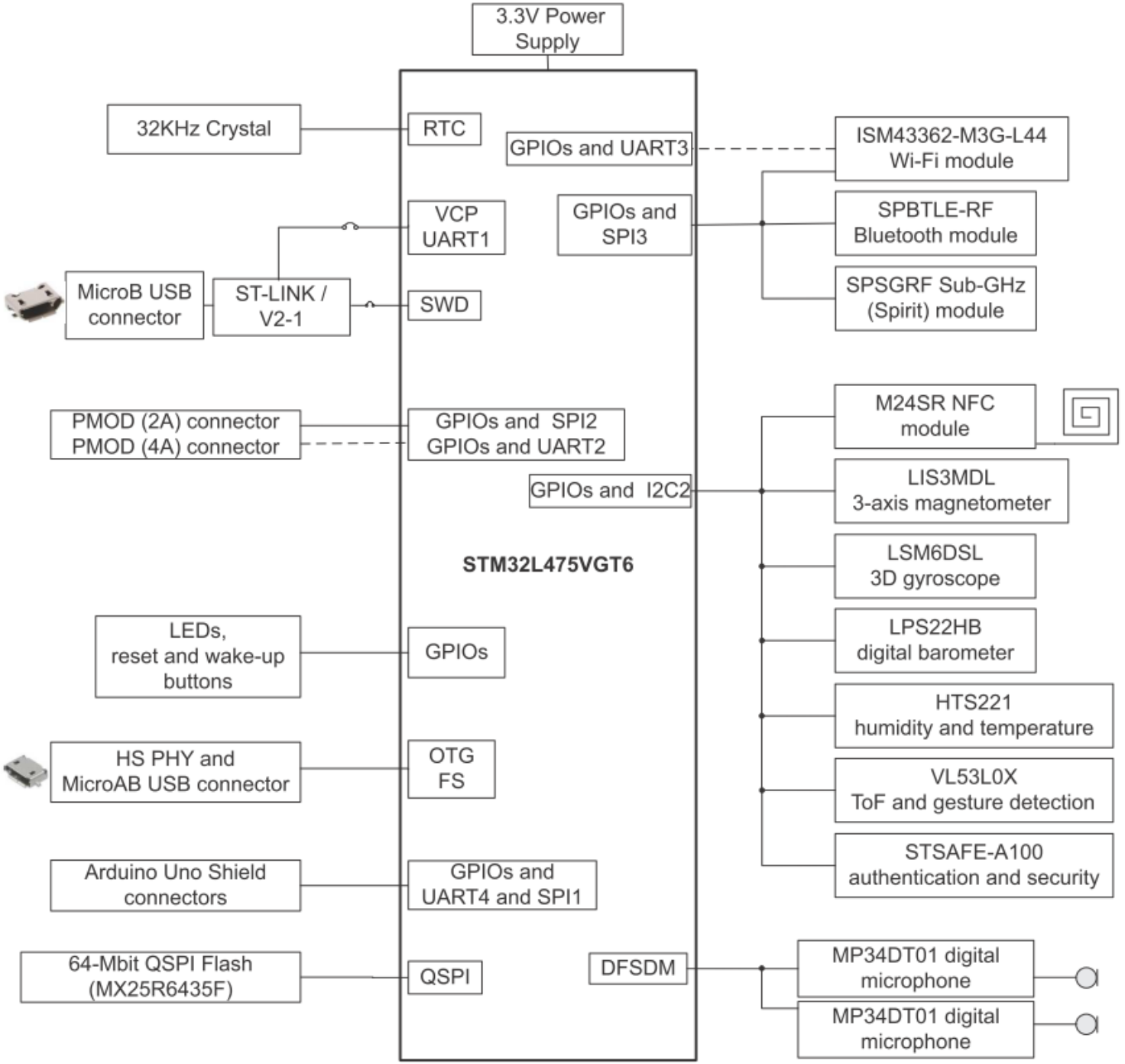
# SensorDemo BLESensor-App Application Description

# B-L475E-IOT01A On board sensors

- Read on board sensors (temperature, humidity, pressure, gyroscope, accelerometer, magnetometer) using ST BSP drivers.



# Keywords

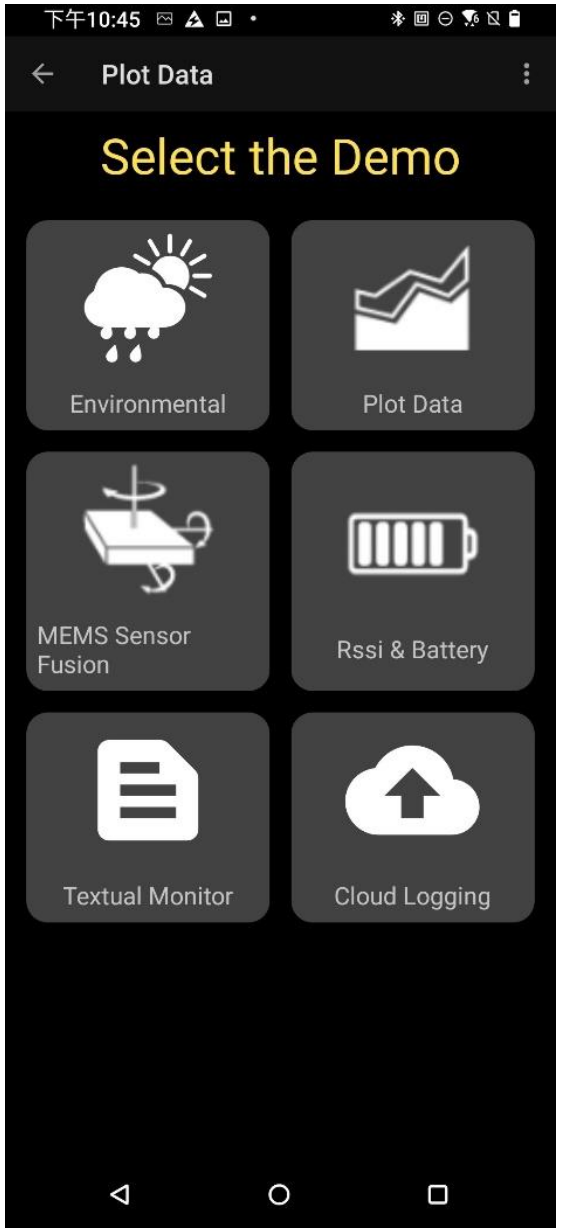
- BlueNRG-MS: One of STM network processor
- BLE:
  - Central/Peripheral
  - GATT client/ GATT server
- Serial: SPI, USART, UART, Virtual Console, I2C
- BSP: Board Support Package
- MSP: MCU Support Package

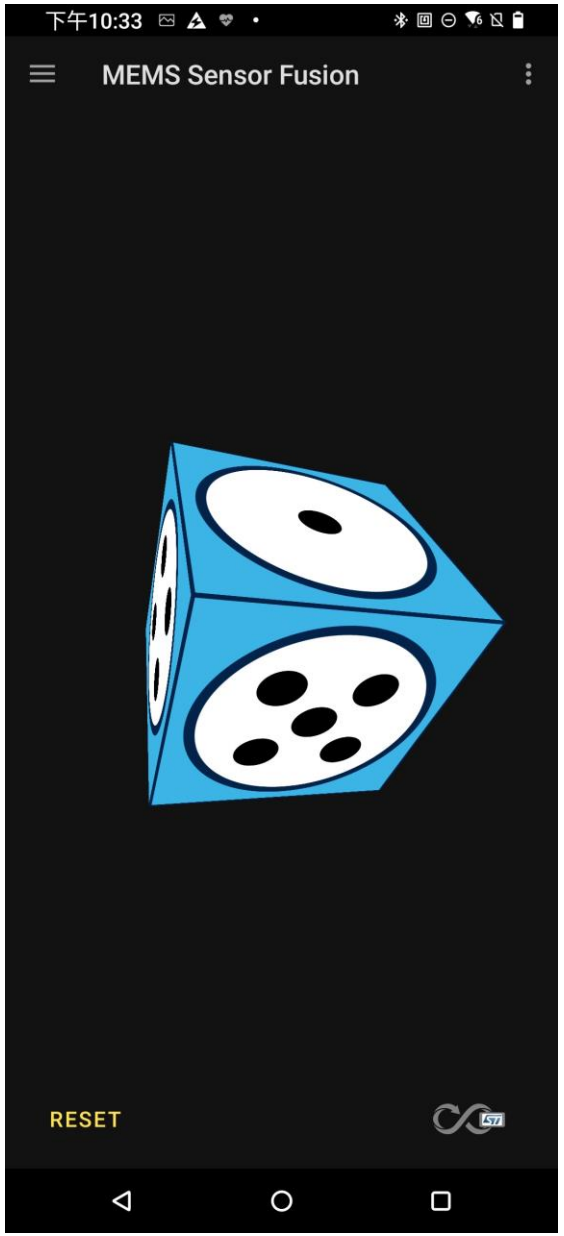
# Demo App Goals

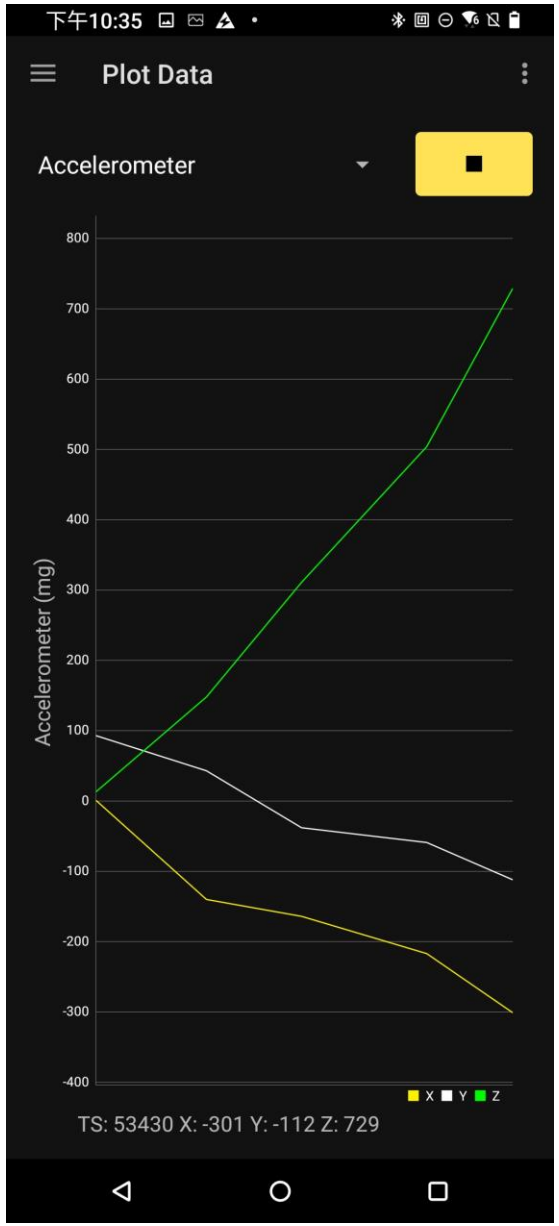
- To implement a peripheral device using STM32 IoT Node
  - Use STM32CubeMX configure the I/O and Software components
  - And generate project codes for STM32CubeIDE
- Testing with the Android/iPhone App: ST BLE Sensor Classic
- After establishing the connection between the STM32 board and the smartphone:

- the temperature and the pressure emulated values are sent by the STM32 board to the mobile device and are shown in the ENVIRONMENTAL tab;
- the emulated sensor fusion data sent by the STM32 board to the mobile device reflects into the cube rotation showed in the app's MEMS SENSOR FUSION tab
- the plot of the emulated data (temperature, pressure, sensor fusion data, accelerometer, gyroscope and magnetometer) sent by the board are shown in the PLOT DATA tab;
- in the RSSI & Battery tab the RSSI value is shown.
- According to the value of the #define USE_BUTTON in file app_bluenrg_ms.c, the environmental and the motion data can be sent automatically (with 1 sec period) or when the User Button is pressed.

# ST BLE Sensor Classic App



# Project overview

- app_bluenrg_ms.c SensorDemo_BLESensor-App initialization and applicative code
- gatt_db.c Functions for building GATT DB and handling GATT events
- hci_tl_interface.c Interface to the BlueNRG HCI Transport Layer
- main.c Main program body
- sensor.c Sensor init and state machine
- stm32**xx_hal_msp.c Source code for MSP (MCU Support Package) Initialization and de-Initialization
- stm32**xx_it.c Source code for interrupt Service Routines
- stm32**xx_board.c Source file for the BSP Common driver
- stm32**xx_board_bus.c Source file for the BSP BUS IO driver
- system_stm32**xx.c CMSIS Cortex-Mx Device Peripheral Access Layer System Source File

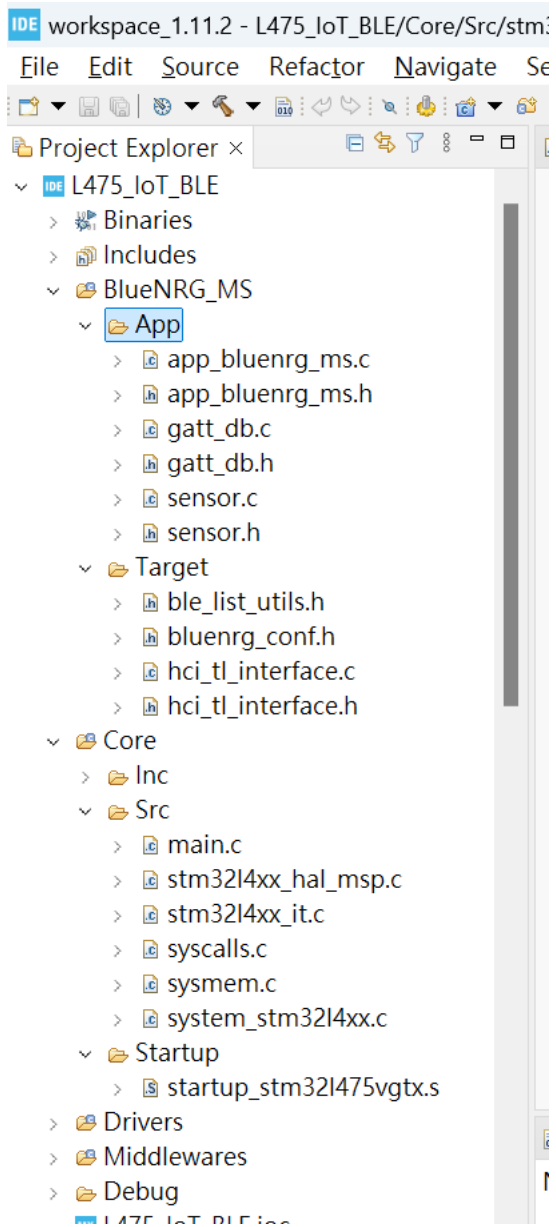
Target dir: Source files for target (HCI) access

# Project structure

- Note if you want to see debug printf message
  - Make sure to have the following “#define” in bluenrg_conf.h

/*---------- Print messages from BLE1 files at user level ----------*/

#define BLE1_DEBUG 1



# app_bluenrg_ms.c: MX_BlueNRG_MS_Init(void) and MX_BlueNRG_MS_Process(void)

- BlueNRG-MS background task - called in the infinite loop of main()

```
void MX_BlueNRG_MS_Process(void)
{
/* USER CODE BEGIN BlueNRG_MS_Process_PreTreatment */

/* USER CODE END BlueNRG_MS_Process_PreTreatment */

User_Process();
hci_user_evt_proc();
/* USER CODE BEGIN BlueNRG_MS_Process_PostTreatment */
/* USER CODE END BlueNRG_MS_Process_PostTreatment */
}
```